

Keplerian Radial-Velocity Fitting for Single-Planet Exoplanet Parameter Estimation

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Abstract

The radial-velocity (RV) method remains one of the foundational techniques for detecting and characterizing extrasolar planets. We present a compact, reproducible pipeline for estimating the orbital parameters of a single companion from Doppler time series. The forward model is the standard Keplerian RV curve parameterized by semi-amplitude K , period P , eccentricity e , argument of periastron ω , time of periastron t_0 , and systemic velocity γ . Parameters are recovered with a global differential-evolution search followed by local nonlinear least-squares refinement. From the fitted amplitude and an assumed stellar mass we derive a minimum planet mass and orbital semi-major axis under the edge-on inclination limit. Synthetic experiments recover planted values of K and P to within a few percent at residual RMS of $\sim 2 \text{ m s}^{-1}$. Application to four public NASA Exoplanet Archive radial-velocity series (51 Peg, HD 209458, 70 Vir, and HD 3651) recovers literature-consistent single-planet solutions, including the short-period nearly circular orbit of 51 Peg b ($P \approx 4.23 \text{ d}$, $K \approx 56.1 \text{ m s}^{-1}$, $M_p \sin i \approx 0.47 M_J$). A velocity-Verlet two-body integration is used only for post-fit orbit visualization.

1 Introduction

The discovery of 51 Pegasi b established Doppler spectroscopy as a practical route to detecting planets around solar-type stars [1, 2]. In the RV method, the star and planet orbit their common barycenter; the line-of-sight component of the stellar motion appears as a periodic Doppler shift in the stellar spectrum. Modern surveys and modeling frameworks treat this signal with analytic Keplerian orbits and statistically motivated fitting procedures [3–5].

For a single planet on a Keplerian orbit, the stellar radial velocity is fully specified by a small set of orbital elements [6, 7]. The semi-amplitude K encodes the planet-to-star mass ratio and orbital scale, while P and e control the periodicity and waveform shape. Because the likelihood surface can be multimodal—especially when eccentricity and phase are free—global optimization or Bayesian sampling is often preferred over purely local gradient methods [8, 9].

This work implements a transparent single-planet RV estimator intended for reproducible analysis and pedagogical demonstration. The pipeline (i) optionally converts Doppler shifts to radial velocities, (ii) estimates a dominant period with a Lomb–Scargle periodogram, (iii) fits a Keplerian model with differential evolution plus least-squares polish, and (iv) derives physical parameters and a barycentric orbit visualization. We validate the method on synthetic data with known ground truth and apply it to four public NASA Exoplanet Archive radial-velocity time series [10].

2 Methods

2.1 Doppler shift and radial velocity

Spectroscopic observations may be reported as a fractional Doppler shift $z = \Delta\lambda/\lambda$. For a source with line-of-sight speed v relative to the observer, the relativistic Doppler formula is

$$z = \sqrt{\frac{1+\beta}{1-\beta}} - 1, \quad \beta = \frac{v}{c}, \quad (1)$$

where c is the speed of light [11]. Inverting this relation yields

$$v = c \frac{(1+z)^2 - 1}{(1+z)^2 + 1}. \quad (2)$$

Throughout, positive v denotes recession.

2.2 Keplerian radial-velocity model

We adopt the standard single-planet RV model [3, 7]:

$$v_r(t) = \gamma + K \left[\cos(\omega + \nu(t)) + e \cos \omega \right], \quad (3)$$

where γ is the systemic velocity, K is the RV semi-amplitude, e is the eccentricity, ω is the argument of periastron, and $\nu(t)$ is the true anomaly.

The true anomaly is obtained from the mean anomaly

$$M = \frac{2\pi}{P} (t - t_0) \quad (4)$$

by solving Kepler's equation

$$E - e \sin E = M \quad (5)$$

for the eccentric anomaly E , then converting

$$\tan \frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{E}{2} \quad (6)$$

[6]. Here P is the orbital period and t_0 is a time of periastron passage. Supplementary notation, unit conventions, and the visualization equations are collected in Appendix A.

For an edge-on orbit ($\sin i = 1$), the semi-amplitude is related to the component masses and period by

$$K = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{M_p}{(M_\star + M_p)^{2/3}} \frac{1}{\sqrt{1-e^2}}, \quad (7)$$

where G is Newton's constant, i is the orbital inclination to the sky plane, $M_\star$ is the stellar mass, and M_p is the planet mass. When evaluating Eq. (7) numerically we convert P to seconds so that the combination is dimensionally consistent in SI units. Given K , P , e , and $M_\star$, we solve Eq. (7) for M_p and obtain the relative orbital semi-major axis from Kepler's third law,

$$a^3 = \frac{G(M_\star + M_p)}{4\pi^2} P^2. \quad (8)$$

When $\sin i < 1$, the mass inferred from Eq. (7) under the edge-on assumption is interpreted as the minimum mass $M_p \sin i$; the general amplitude formula appears in Appendix A.

2.3 Parameter estimation

Let $\{t_j, v_j, \sigma_j\}$ denote the observed times, radial velocities, and uncertainties. We minimize the weighted residual sum of squares

$$\chi^2(\boldsymbol{\theta}) = \sum_j \left(\frac{v_j - v_r(t_j; \boldsymbol{\theta})}{\sigma_j} \right)^2, \quad (9)$$

with parameter vector $\boldsymbol{\theta} = (K, P, e, \omega, t_0, \gamma)$.

Because χ^2 is nonconvex in the orbital elements, we first estimate a dominant period with a normalized Lomb–Scargle periodogram [12] and then perform a global search with differential evolution [9, 12], which maintains a population of candidate solutions and updates them by mutation and crossover. The best candidate is then polished with a bounded nonlinear least-squares solver [12]. For the observational series the period search window is centered on the periodogram peak; eccentricity is restricted to $0 \leq e \leq 0.85$ to avoid near-parabolic degeneracies. Numerical settings and per-series bounds are listed in Appendix D. For the synthetic validation run we also record the best-so-far χ^2 after each differential-evolution generation (Figure 6).

2.4 Orbit visualization

After fitting, we construct barycentric initial conditions consistent with the derived a , e , and component masses, and integrate the Newtonian two-body problem with the velocity Verlet algorithm [13]. This integration is used solely for visualization of the reflex orbit; it is not part of the likelihood evaluation. The explicit initial-condition and update equations are given in Appendix A.

3 Data

3.1 Synthetic validation set

To quantify recovery accuracy we generate a synthetic RV series for a Jupiter-class companion around a $\sim 1.08 M_\odot$ star with planted parameters

$$\begin{aligned} P &= 147.5 \text{ d}, & K &\approx 30.5 \text{ m s}^{-1}, & e &= 0.19, \\ \omega &= 1.05 \text{ rad}, & t_0 &= 18.0 \text{ d}, & \gamma &= 4.5 \text{ m s}^{-1}, & M_p &= 0.82 M_J. \end{aligned}$$

Observation times are drawn irregularly over a 420 d window. Independent Gaussian measurement noise of standard deviation $\sigma_n = 2.2 \text{ m s}^{-1}$ and an independent Gaussian jitter draw of standard deviation $\sigma_{\text{jit}} = 0.7 \text{ m s}^{-1}$ are added to each sample (Appendix A).

3.2 Public NASA Exoplanet Archive series

We analyze four published radial-velocity curves served by the NASA Exoplanet Archive [10]:

- 51 Peg [14], archive file UID_0113357_RVC_002.tb1;
- HD 209458 [15], archive file UID_0108859_RVC_001.tb1;
- 70 Vir [2, 15], archive file UID_0065721_RVC_001.tb1;
- HD 3651 [15], archive file UID_0003093_RVC_002.tb1.

Each series provides Julian dates, barycentric radial velocities in m s^{-1} , and published per-point uncertainties. Host masses (1.07, 1.23, 1.09, and $0.799 M_{\odot}$) are taken from the archive Planetary Systems default solutions. For fitting we shift each time axis so that the first observation is at $t = 0$, leaving the relative cadence unchanged. Working CSV copies and a machine-readable provenance catalog are packaged under `data/nasa_exoplanet_archive/`.

Data provenance. All observational inputs used here are public NASA Exoplanet Archive RVC products downloaded from the archive bulk/time-series service. Original IPAC table headers retain the literature reference, telescope, and instrument metadata; those identifiers are preserved in `catalog.json` so each fit can be traced to a specific archive file and paper.

4 Results

4.1 Synthetic recovery

Table 1 compares planted and recovered parameters. The period and semi-amplitude are recovered to within $\sim 3\%$, and the residual RMS (Appendix A) is 2.11 m s^{-1} , consistent with the injected noise level. Figure 1 shows the observations, the best-fit Keplerian curve, the ground-truth curve, and the corresponding barycentric orbit.

Table 1: Synthetic validation: planted truth versus recovered fit.

Quantity	Truth	Fit	Relative error
P (d)	147.5	147.16	0.2%
K (m s^{-1})	30.51	31.35	2.8%
e	0.190	0.221	—
M_p (M_J)	0.820	0.836	2.0%
RMS (m s^{-1})	—	2.11	—

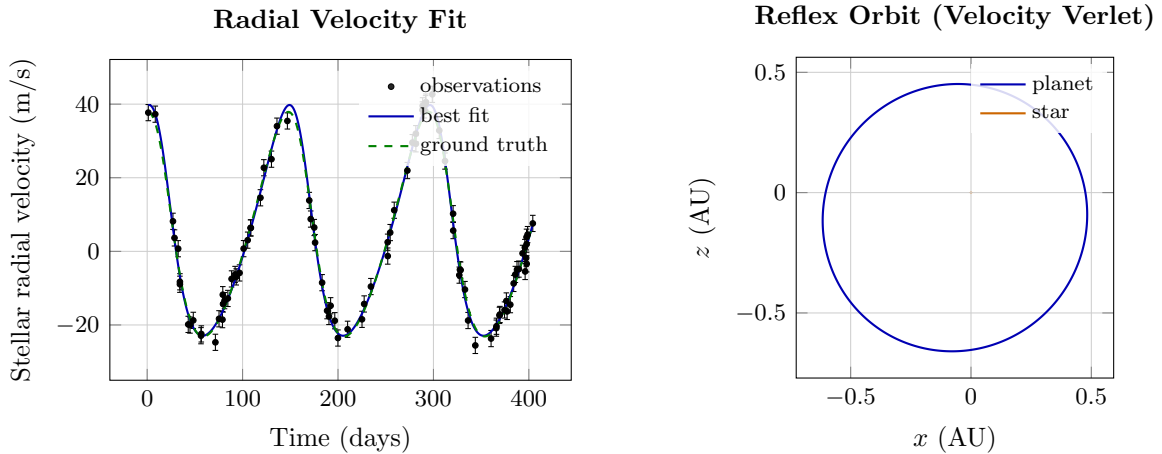


Figure 1: Synthetic radial-velocity fit. Left: noisy observations (points), best-fit Keplerian model (solid), and planted ground truth (dashed). Right: barycentric star-planet orbit integrated with velocity Verlet from the fitted elements.

4.2 Observational fits for public archive series

Table 2 summarizes the best-fit Keplerian parameters for all four public series. Figure 2 shows the 51 Peg recovery; the remaining RV and orbit panels appear in Appendix B, and the full six-element solutions are collected in Appendix C.

51 Peg yields a short-period, nearly circular solution with residual RMS 5.20 m s^{-1} . Under the edge-on assumption the companion mass is $M_p \sin i \approx 0.47 M_J$ at $a \approx 0.052 \text{ AU}$, consistent with the classic hot-Jupiter architecture [1, 14].

HD 209458 likewise recovers a short, nearly circular orbit ($P \approx 3.525 \text{ d}$, $K \approx 84.4 \text{ m s}^{-1}$) with residual RMS 4.78 m s^{-1} . 70 Vir and HD 3651 recover longer-period eccentric solutions ($P \approx 116.7$ and 62.2 d ; $e \approx 0.40$ and 0.65) with residual RMS of 7.1 and 7.0 m s^{-1} , respectively.

Table 2: Best-fit single-planet Keplerian parameters for the four public NASA Exoplanet Archive RV series. Masses assume $\sin i = 1$. Full orbital elements are given in Table 4.

Quantity	51 Peg	HD 209458	70 Vir	HD 3651
N points	111	64	74	121
$M_\star (M_\odot)$	1.07	1.23	1.09	0.799
P (d)	4.231	3.525	116.688	62.197
K (m s^{-1})	56.05	84.35	316.8	16.31
e	0.015	0.012	0.402	0.648
γ (m s^{-1})	7.33	0.05	-0.18	-0.33
a (AU)	0.052	0.049	0.482	0.285
$M_p \sin i (M_J)$	0.467	0.725	7.42	0.209
RMS (m s^{-1})	5.20	4.78	7.08	7.04

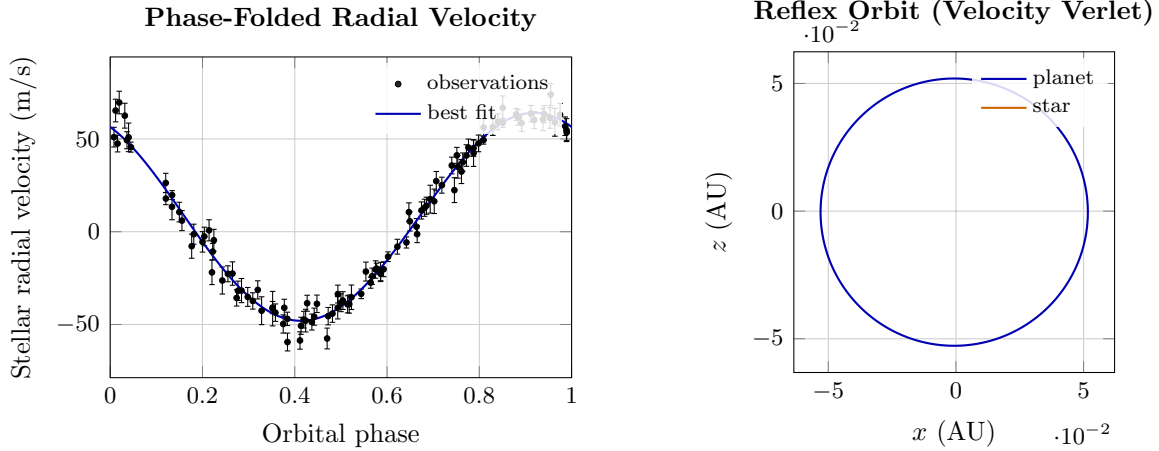


Figure 2: Phase-folded Keplerian fit to the public 51 Peg radial-velocity series of Marcy et al. [14]. Left: observations folded on the fitted period with the best-fit model. Right: corresponding barycentric orbit. Fits for HD 209458, 70 Vir, and HD 3651 are shown in Appendix B.

5 Discussion

The synthetic experiment confirms that differential evolution followed by local refinement is sufficient to recover the dominant Keplerian parameters when the noise is Gaussian and the sampling covers multiple orbital cycles [3, 9]. Eccentricity is the most weakly constrained element in our synthetic run, as expected when the signal-to-noise ratio is modest and the true e is moderate [8].

Across the observational sample, 51 Peg and HD 209458 behave like compact, low-eccentricity hot-Jupiter signals of the kind first revealed by early Doppler surveys [1, 14], while 70 Vir and HD 3651 illustrate longer-period eccentric recoveries with acceptable residuals [2, 15]. Several caveats apply to all four fits. First, although published uncertainties are used for weighting, residual systematics and stellar jitter are not modeled explicitly, so formal χ^2 values should not be over-interpreted. Second, the masses we report are $M_p \sin i$; without an independent inclination constraint the true masses may be larger. Third, the model assumes a single planet and a constant systemic velocity, omitting stellar activity, instrumental offsets between instruments, and additional companions [5]. The present solutions are therefore best viewed as reproducible single-Keplerian pipeline validation on named public archive series rather than as replacement orbital ephemerides.

Natural extensions include periodogram-based period initialization [4], multi-planet Keplerian models [3], and Bayesian posterior sampling for full uncertainty quantification [8].

6 Conclusions

We have described a Keplerian radial-velocity pipeline for single-planet parameter estimation. The method combines an analytic RV forward model, Lomb–Scargle period initialization, differential-evolution global search, and least-squares refinement, with derived planet mass and semi-major axis under the edge-on limit. Synthetic tests recover K and P to within a few percent. Application to four public NASA Exoplanet Archive series recovers literature-consistent short-period circular solutions for 51 Peg and HD 209458 together with longer-period eccentric solutions for 70 Vir and HD 3651. The accompanying open-source implementation provides a reproducible basis for further methodological development.

Acknowledgments

This research has made use of the NASA Exoplanet Archive, which is operated by the California Institute of Technology, under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program [10]. The radial-velocity time series analyzed here are the public archive products of Marcy et al. [14] and Butler et al. [15].

Software

The analysis is implemented in Python using NumPy, SciPy [12], and Matplotlib. Source code and example scripts are available in the project repository accompanying this manuscript. Commands used to regenerate the paper figures are listed in Appendix E.

A Notation and supplementary definitions

Symbols

Unless noted otherwise, times are expressed in days in the fitting interface and converted to seconds only when evaluating Newtonian formulas that involve G . Masses $M_\odot$ and M_J denote one solar mass and one Jupiter mass, respectively. The Doppler observables use rest wavelength λ and observed wavelength shift $\Delta\lambda$, so that $z = \Delta\lambda/\lambda$. The fitted velocities v_j are identified with the line-of-sight speed v obtained from z .

The orbital inclination i is the angle between the orbital angular-momentum vector and the line of sight, so that an edge-on orbit has $i = 90^\circ$ and $\sin i = 1$. The relative orbital semi-major axis a is the semi-major axis of the planet–star separation vector.

General semi-amplitude

Allowing for arbitrary inclination, Eq. (7) generalizes to

$$K = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{M_p \sin i}{(M_\star + M_p)^{2/3}} \frac{1}{\sqrt{1 - e^2}}, \quad (10)$$

with P in seconds when G is in SI units. Inverting Eq. (7) with $\sin i = 1$ therefore returns $M_p \sin i$ whenever the true inclination is unknown.

Residual diagnostics

For N observations the unweighted residual root-mean-square reported in the tables is

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{j=1}^N (v_j - v_r(t_j; \hat{\boldsymbol{\theta}}))^2}, \quad (11)$$

where $\hat{\boldsymbol{\theta}}$ is the best-fit parameter vector. This is distinct from the weighted objective $\chi^2(\boldsymbol{\theta})$ used during optimization. Relative errors in validation tables are $|\hat{q} - q_{\text{true}}|/|q_{\text{true}}|$ for a scalar quantity q .

Synthetic noise model

Synthetic observations are generated as

$$v_j = v_r(t_j; \boldsymbol{\theta}_{\text{true}}) + \varepsilon_{n,j} + \varepsilon_{\text{jit},j}, \quad (12)$$

where $\varepsilon_{n,j} \sim \mathcal{N}(0, \sigma_n^2)$ and $\varepsilon_{\text{jit},j} \sim \mathcal{N}(0, \sigma_{\text{jit}}^2)$ are independent. The fitting weights use $\sigma_j = \sigma_n$ and therefore do not include the jitter variance. For the public archive series we use the published per-point uncertainties supplied with each RVC file.

Orbit visualization

Orbit panels show barycentric trajectories in an orbital x – z plane with the line-of-sight direction along $+z$. Given relative polar radius

$$r = \frac{a(1 - e^2)}{1 + e \cos \nu} \quad (13)$$

and argument of periastron ω , the planet–star relative state in that plane is obtained by rotating the perifocal vectors through ω . Barycentric positions and velocities then follow from the mass ratio $\mu_p = M_p/(M_\star + M_p)$,

$$\mathbf{r}_\star = -\mu_p \mathbf{r}, \quad \mathbf{r}_p = (1 - \mu_p) \mathbf{r}, \quad (14)$$

and likewise for the velocities. Starting from those initial conditions, the Newtonian accelerations

$$\ddot{\mathbf{r}}_\star = GM_p \frac{\mathbf{r}_p - \mathbf{r}_\star}{|\mathbf{r}_p - \mathbf{r}_\star|^3}, \quad \ddot{\mathbf{r}}_p = -GM_\star \frac{\mathbf{r}_p - \mathbf{r}_\star}{|\mathbf{r}_p - \mathbf{r}_\star|^3} \quad (15)$$

are integrated with the velocity Verlet update

$$\begin{aligned} \mathbf{r}(t + \Delta t) &= \mathbf{r}(t) + \mathbf{v}(t) \Delta t + \frac{1}{2} \mathbf{a}(t) (\Delta t)^2, \\ \mathbf{v}(t + \Delta t) &= \mathbf{v}(t) + \frac{1}{2} (\mathbf{a}(t) + \mathbf{a}(t + \Delta t)) \Delta t. \end{aligned} \quad (16)$$

The visualization uses a fixed number of steps over one fitted orbital period and does not enter the RV likelihood.

B Additional observational figures

Figures 3–5 show the RV fits and barycentric orbits for the three additional public archive series discussed in the main text.

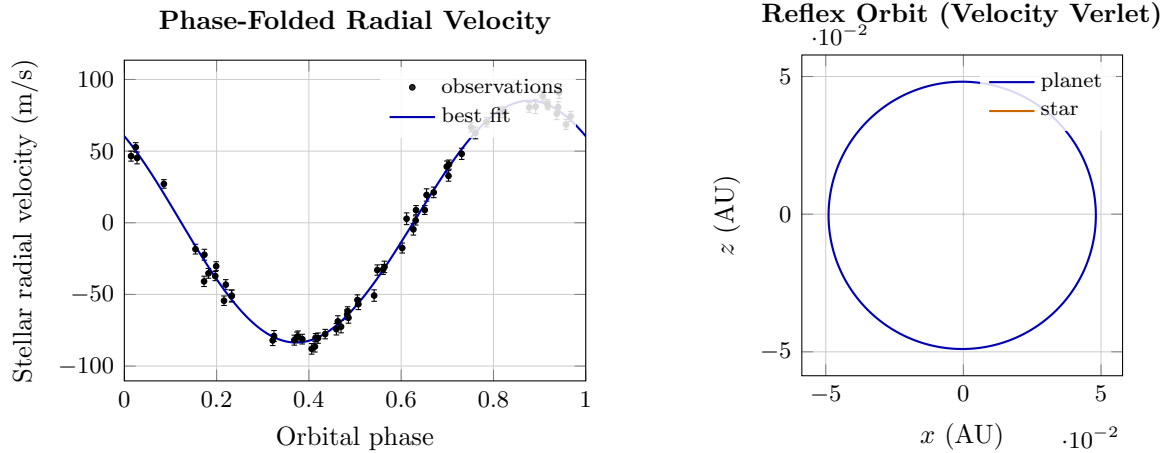


Figure 3: Phase-folded Keplerian fit to the public HD 209458 radial-velocity series of Butler et al. [15].

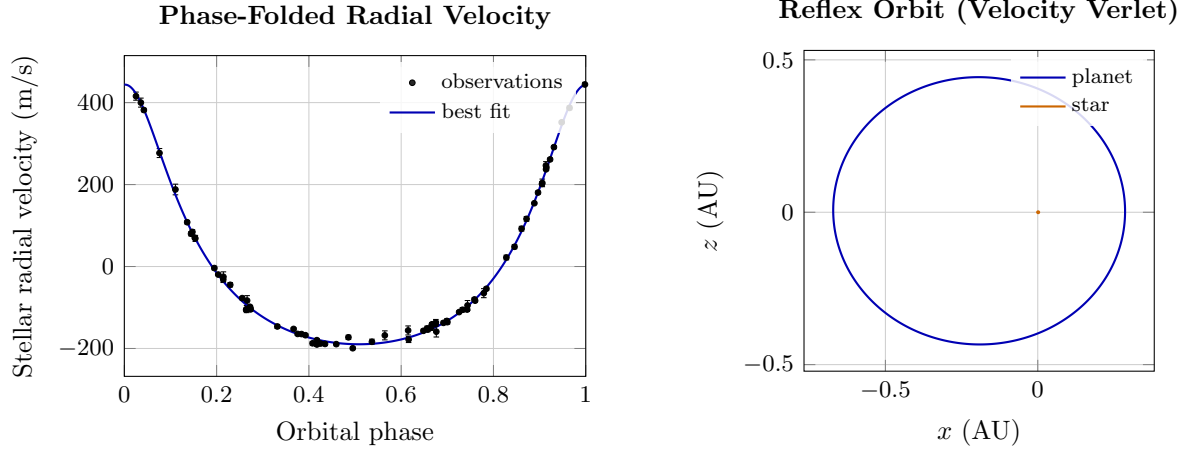


Figure 4: Phase-folded Keplerian fit to the public 70 Vir radial-velocity series of Butler et al. [15].

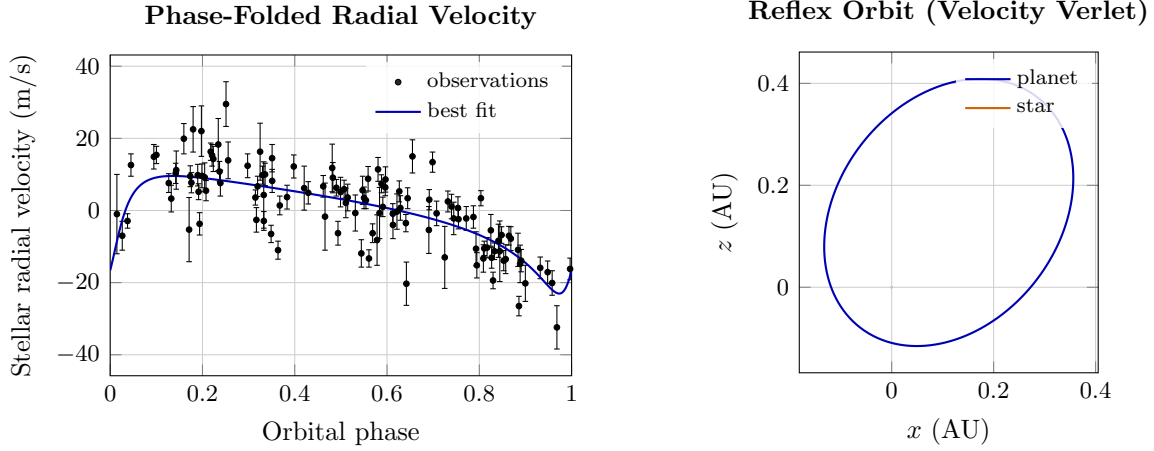


Figure 5: Phase-folded Keplerian fit to the public HD 3651 radial-velocity series of Butler et al. [15].

C Full Keplerian solutions

Table 3 reports all six fitted Keplerian elements for the synthetic validation run. Table 4 lists the corresponding solutions for the four public archive series. Times of periastron are reported in days relative to the first observation of each series and need not fall inside the observational window because the mean anomaly is 2π -periodic.

Table 3: Full synthetic Keplerian solution (planted truth versus recovered fit).

Parameter	Truth	Fit	Relative error
K (m s^{-1})	30.51	31.35	2.8%
P (d)	147.5	147.16	0.2%
e	0.190	0.221	—
ω (rad)	1.050	1.000	—
t_0 (d)	18.0	459.0	—
γ (m s^{-1})	4.50	4.68	—
M_p (M_J)	0.820	0.836	2.0%
a (AU)	—	0.560	—
RMS (m s^{-1})	—	2.11	—
χ^2	—	84.9	—

Table 4: Full single-planet Keplerian solutions for the public archive series. Masses assume $\sin i = 1$.

Parameter	51 Peg	HD 209458	70 Vir	HD 3651
K (m s^{-1})	56.05	84.35	316.8	16.31
P (d)	4.231	3.525	116.688	62.197
e	0.0151	0.0125	0.402	0.648
ω (rad)	0.529	0.786	-0.025	-2.222
t_0 (d)	-9.950	1.161	44.8	42.0
γ (m s^{-1})	7.33	0.05	-0.18	-0.33
a (AU)	0.0524	0.0486	0.482	0.285
$M_p \sin i$ (M_J)	0.467	0.725	7.42	0.209
RMS (m s^{-1})	5.20	4.78	7.08	7.04
χ^2	108.2	112.4	120.1	516.7

D Optimizer configuration

The global stage uses SciPy’s differential-evolution routine with `polish=False`, `updating=deferred`, a fixed random seed, and at most 300 iterations for the synthetic run (400 for the observational series). The local stage is SciPy’s bounded nonlinear least-squares solver applied to the weighted residual vector, with a budget of 2×10^4 function evaluations. The free parameter vector is ordered as $(K, P, e, \omega, t_0, \gamma)$.

Figure 6 shows the synthetic-run convergence trace: the best population χ^2 after each differential-evolution generation, the final least-squares polish, and the relative errors in K and P against the planted truth. Early generations reduce χ^2 by orders of magnitude; once P is approximately recovered, further improvement is comparatively gradual before the local polish.

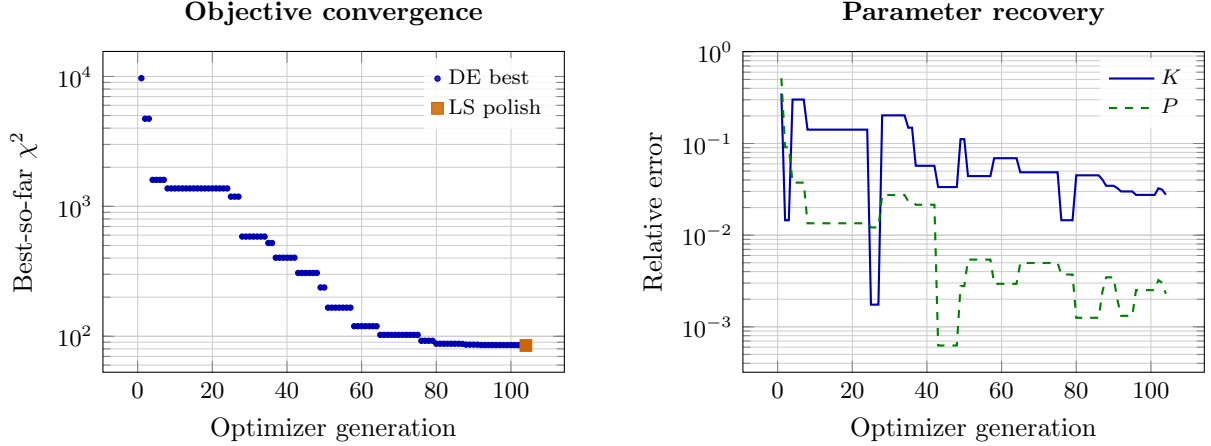


Figure 6: Synthetic optimizer convergence. Left: best-so-far weighted χ^2 after each differential-evolution generation (circles) and after the final least-squares polish (square). Right: relative errors in the recovered semi-amplitude K and period P versus the planted truth.

For the synthetic dataset the default bound construction of the package is used. For the public archive series we center the period window on the Lomb–Scargle peak and adopt $0 \leq e \leq 0.85$ with $-\pi \leq \omega \leq \pi$. Table 5 lists the numerical bounds used for each fit.

Table 5: Search bounds used for each fit. Units are ms^{-1} for K and γ , days for P and t_0 , and radians for ω .

Series	K	P	e	ω	t_0	γ
Synthetic	[1, 102]	[50.4, 604]	[0, 0.85]	$[-\pi, \pi]$	$[-603, 606]$	$[-107, 97]$
51 Peg	[1, 267]	[1.41, 12.7]	[0, 0.85]	$[-\pi, \pi]$	$[-12.7, 12.7]$	$[-266, 267]$
HD 209458	[1, 358]	[0.5, 4.17]	[0, 0.85]	$[-\pi, \pi]$	$[-4.2, 4.2]$	$[-378, 337]$
70 Vir	[1, 1287]	[38.9, 350]	[0, 0.85]	$[-\pi, \pi]$	$[-350, 350]$	$[-1377, 1198]$
HD 3651	[1, 124]	[20.8, 187]	[0, 0.85]	$[-\pi, \pi]$	$[-187, 187]$	$[-122, 126]$

E Reproducibility commands

From the project root, regenerate the CSV tables and light-theme vector PDFs, then compile this manuscript, with

```
MPLBACKEND=Agg uv run python examples/export_paper_figures.py
latexmk -pdf -cd docs/exoplanet_showcase_report.tex
```

Print figures in the PDF are drawn natively with PGFplots from the exported CSV tables under `docs/figures/plotdata/`. The same export script also writes Matplotlib PDF copies under `docs/figures/`.

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